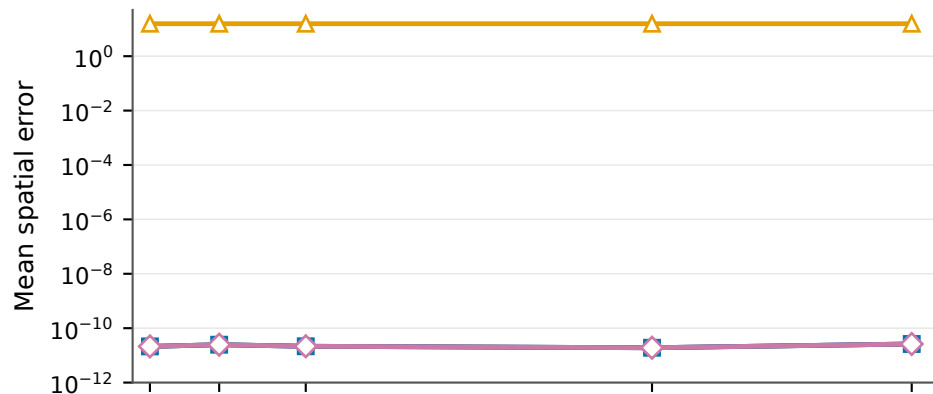


Alignment residuals across input rotations

Each marker is one validated run. Colours denote simulated conditions; no cross-condition aggregation or method ranking. Mean spatial-error panels share a log scale; rotation-error rows use their own raw linear limits.

Baseline Mean $\times 0.5$ Variance $\times 2.0$ Sparsity $\times 0.5$

Mean spatial error $\downarrow$ (log scale)



Rotation recovery error (deg) $\downarrow$

